Goal

Grow your organism to become the largest!



Rules

The game is played on a grid.

The Organisms

Organisms are made up of **organs** that take up one tile of space on the game grid.

Each player starts with a ROOT type organ. Your organism can GROW a new organ on each turn in order to cover a larger area.

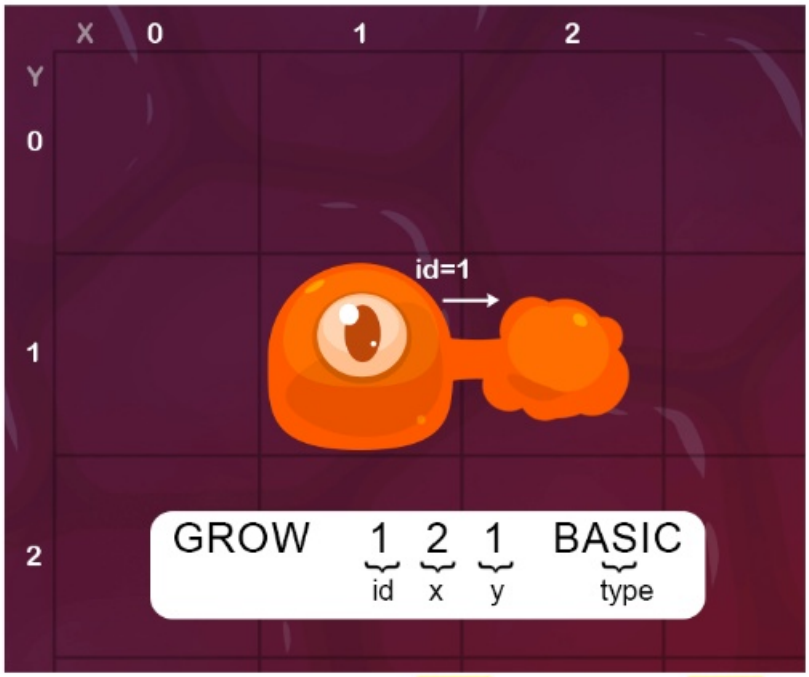
A new organ can grow **from** any existing **organ**, onto an **empty adjacent location**.

In order to GROW, your organism needs **proteins**. Growing 1 BASIC organ requires 1 protein of type A.

You can obtain more **proteins** by growing an organ onto a tile of the grid containing a **protein source**, these are tiles with a letter in them. Doing so will grant you 3 proteins of the corresponding type.

You organism can receive the following command:

- `GROW id x y type`: creates a new organ at location `x`, `y` from organ with id `id`. If the target location is not a neighbour of `id`, the organ will be created on the shortest path to `x`, `y`.



This command will create new `BASIC` organ with the `ROOT` organ as its parent.

See the **Game Protocol** section for more information on sending commands to your organism.

HARVESTER Rules

Organs you place may be given a `direction`.



This command will create new `HARVESTER` facing `N` (North).

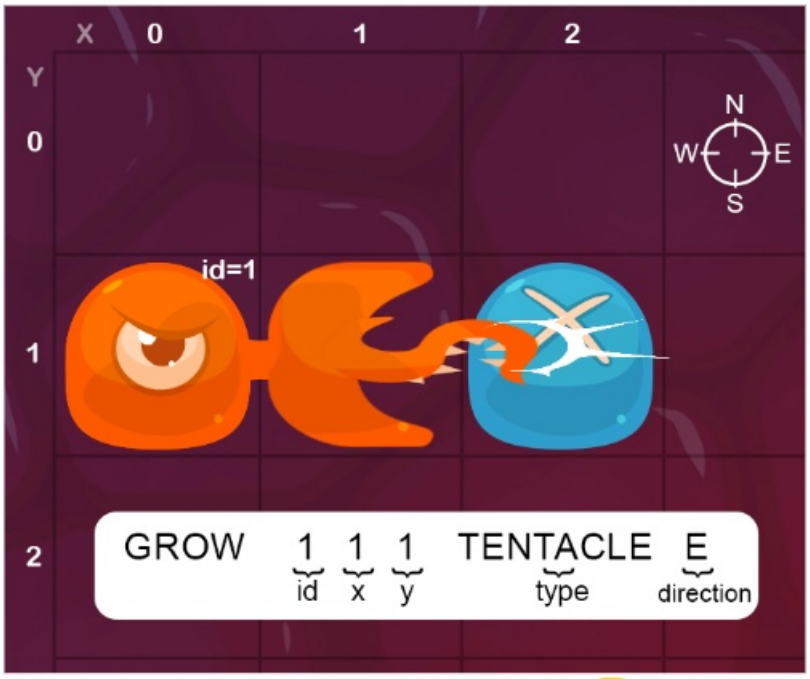
If a `HARVESTER` is facing a tile with a **protein source**, you will receive `1` of that protein on **every end of turn**.

Note: each player gains only `1` protein from each source per turn, even if multiple harvesters are facing that source.

To grow a HARVESTER you need 1 C type protein and 1 D type protein.

TENTACLERules

On each turn, right after **harvesting** , any TENTACLE organs facing an opponent organ will **attack** , causing the target organ to **die**. Attacks happen simultaneously.



This command will create a new TENTACLE facing E (East), causing the opponent organ to be attacked.

When an organ **dies,** all of its **children** also die. This will propagate to the **entire** organism if the ROOT is hit.

Note: You can use the organParentId variable to keep track of each organ's children.

A tentacle also prevents the opponent from growing onto the tile it is facing.

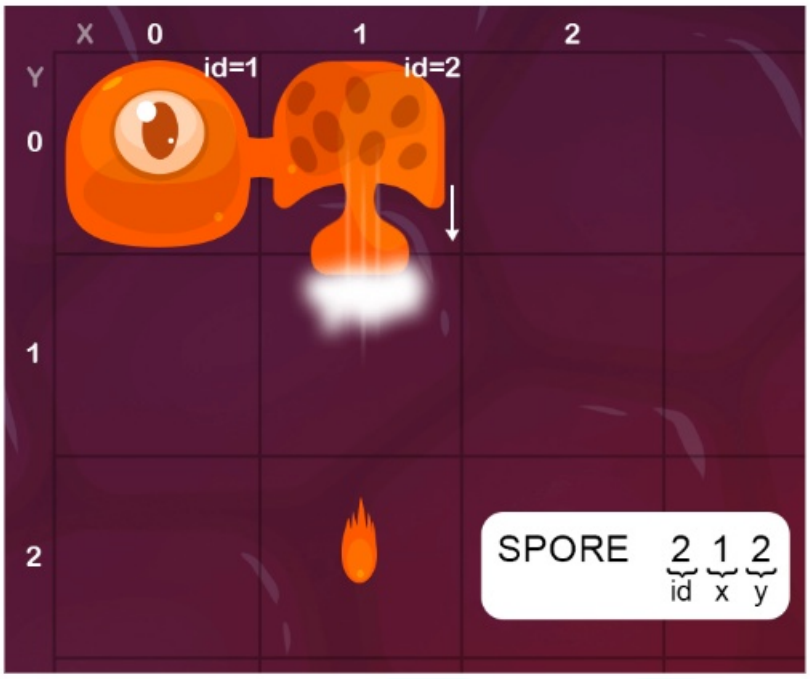
To grow a TENTACLE you need 1 B type protein and 1 C type protein.

SPORERRules

The SPORER type organ is unique in two ways:

- It is the only organ that can create a new ROOT organ.
- To create a new ROOT, it shoots out a spore in a straight line, letting you place the new organ in any of the free spaces it is facing.

*Note: a ROOT organ never has a **parent**, even when spawned from a SPORER.*



This command will make the `SPORER` *shoot a new* `ROOT` *to the South.*

When you control **multiple** organisms, you must output one command for **each one**. They will perform their actions simultaneously.

The `requiredActionsCount` variable will keep track of how many organisms you have. You **must** use the `WAIT` command for any organism that cannot act.

Note: You can use the `organRootId` *variable to find out which organs belong to the same organism.*

To grow a `SPORER` you need `1` `B` type protein and `1` `D` type protein.

To spore a new `ROOT` you need `1` of each protein.

Here is a table to summarize all organ costs:

Organ	A	B	C	D
`BASIC`	1	0	0	0
`HARVESTER`	0	0	1	1
`TENTACLE`	0	1	1	0
`SPORER`	0	1	0	1
`ROOT`	1	1	1	1

⛔ Game end

The game stops when it detects progress can no longer be made or after `100` turns.

🎬 Action order for one turn

1. `GROW` and `SPORE` actions are computed.

2. Extra walls from growth collisions are spawned.
3. Protein harvests are computed.
4. Tentacle attacks are computed.
5. Game over conditions are checked.

Victory Conditions

The winner is the player with the most tiles occupied by one of their organs.

Defeat Conditions

Your program does not provide a command in the alloted time or one of the commands is invalid.

Debugging tips

- Hover over the grid to see extra information on the organ under your mouse.
- Append text after any command and that text will appear above your organism.
- Press the gear icon on the viewer to access extra display options.
- Use the keyboard to control the action: space to play/pause, arrows to step 1 frame at a time.

Game Protocol

Initialization Input

First line: two integers `width` and `height` for the size of the grid.

Input for One Game Turn

First line: one integer `entityCount` for the number of entities on the grid.
Next `entityCount` **lines:** the following `7` inputs for each entity:

- `x`: X coordinate (`0` is leftmost)
- `y`: Y coordinate (`0` is topmost)
- `type`:
  - `WALL` for a wall
  - `ROOT` for a ROOT type organ
  - `BASIC` for a BASIC type organ
  - `HARVESTER` for a HARVESTER type organ
  - `TENTACLE` for a TENTACLE type organ
  - `SPORER` for a SPORER type organ
  - `A` for an A protein source
  - `B` for a B protein source
  - `C` for a C protein source
  - `D` for a D protein source
- `owner`:
  - `1` if you are the owner of this organ

- `0` if your opponent owns this organ
- `-1` if this is not an organ

- `organId`: unique id of this entity if it is an organ, `0` otherwise
- `organDir`: `N`, `W`, `S`, or `E` for the direction in which this organ is facing
- `organParentId`: if it is an organ, the `organId` of the organ that this organ grew from (0 for `ROOT` organs), else `0`.
- `organRootId`: if it is an organ, the `organId` of the `ROOT` that this organ originally grew from, else `0`.

Next line: `4` integers: `myA`, `myB`, `myC`, `myD` for the amount of each protein type you have.
Next line: `4` integers: `oppA`, `oppB`, `oppC`, `oppD` for the amount of each protein type your opponent has.
Next line: the integer `requiredActionsCount` which equals the number of command you have to perform during the turn.

Output

A single line per organism with its action:

- `GROW id x y type direction` : attempt to grow a new organ of type `type` at location `x`, `y` from organ with id `id`. If the target location is not a neighbour of `id`, the organ will be created on the shortest path to `x`, `y`.
- `SPORE id x y` : attempt to create a new `ROOT` organ at location `x`, `y` from the `SPORER` with id `id`.
- `WAIT` : do nothing.

Append text to your command and it will be displayed in the viewer.

Constraints

Response time per turn ≤ `50` ms
Response time for the first turn ≤ `1000` ms
`16` ≤ `width` ≤ `24`
`8` ≤ `height` ≤ `12`